

AVIK CHAKRAVARTY

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EDUCATION

University of Pennsylvania, Philadelphia, PA.

PhD, Mathematics (Advisors: Angela Gibney & Tony Pantev)

Aug. 2022 - Present

Master of Arts, Mathematics, Bridge-to-PhD program

Aug. 2020 - May 2022

RESEARCH INTERESTS

Mirror symmetry, logarithmic geometry, moduli of curves, and geometric Langlands.

PREPRINTS

Positivity of coinvariant divisors on $\overline{M}_{0,n}$ and the Parafermions

<https://arxiv.org/abs/2506.17593>

Abstract: We give criteria for determining the positivity of line bundles coming from vertex operator algebras (VOAs) on the moduli space $\overline{M}_{0,n}$ of rational curves with n marked points. The criteria use the multiplicative structure of VOA representations encoded in the fusion ring. Using them, we construct positive line bundles on $\overline{M}_{0,n}$ from certain parafermion VOAs. These give the first examples of commutant VOAs producing positive line bundles.

SELECTED TALKS

Positivity of line bundles from VOA representations

Geometry & Topology seminar, Virginia Tech.

Nov. 2025

Coinvariant divisors on $\overline{M}_{0,n}$ and the Parafermions

Math/Phys seminar, University of Pennsylvania.

Sept. 2025

Virtual fundamental classes

eCHT seminar, Wayne State University.

Aug. 2024

Conformal field theory and geometric Langlands

Algebraic geometry student research seminar, University of Pennsylvania.

Mar. 2024

SEMINARS

Organizer of [Basic notions seminar in Math/Physics](#)

Fall 2025

Organizer of [Fukaya categories and Mirror symmetry](#)

Summer 2025

Organizer of [Mirror symmetry and Math/Physics student seminar](#)

Spring 2025

Co-organizer of [Graduate algebraic geometry student seminar](#)

Fall 2024, Sprint 2025

SELECTED WORKSHOPS

Homological Mirror symmetry

IMSA, Miami, FL.

Jan. 2026

FRG Fall school on Quantizations and Lagrangians

University of Illinois, Chicago, IL.

Sep. 2025

Bootcamp & 2025 Summer research institute in Algebraic geometry

Colorado State University, Fort Collins, CO.

Jul. 2025

Summer school: Enumerative geometry and Geometric rep. theory

Università di Pisa, Italy.

Jun. 2025

Homological Mirror symmetry

IMSA, Miami, FL.

Jan. 2025

Winter school on Enumerative geometry and Mirror symmetry

CalTech, Pasadena, CA.

*Jan. 2025***Logs and stacks in birational geometry and moduli**

LMS Invited Lecture Series 2024, London.

*Jul. 2024***A panorama of moduli spaces**

Goethe University Frankfurt am Main, Frankfurt, Germany.

*Feb. 2024***Arithmetic, birational geometry and moduli spaces**

Brown university, Providence, RI.

*Jun. 2023***TEACHING EXPERIENCE**

Instructor for Math 3120: Linear algebra, University of Pennsylvania*Summer 2022.*

This course aims to introduce various proof techniques to study vector spaces and linear maps abstractly to a group of non-math majors. The class covers vector spaces, dual vector spaces, determinant forms, linear maps, eigenvalues and eigenvectors, diagonalizability, Jordan normal forms. Responsibilities include giving lectures, holding office hours and grading.

Teaching Assistant for Math 1400: Multivariable calculus, University of Pennsylvania*Fall 2021.*

This course aims to introduce various aspects of multi-variable calculus to a group of undergraduate students with various majors. Primary responsibility includes conducting weekly recitation sections, office hours and grading weekly quizzes.

Teaching Assistant for Math 3140: Adv. Linear algebra, University of Pennsylvania*Fall 2024, Fall 2023, Spring 2022.*

This course aims to teach a mix of math and non-math majors how to prove central results about vector spaces, linear maps and inner product spaces. Primary responsibility includes conducting weekly recitation sections, office hours and grading quizzes, homework and exams.

Teaching Assistant for Math 2030: Proofs in algebra, University of Pennsylvania*Spring 2024.*

This course aims to introduce basic proof techniques useful in algebra to an audience of math-major and non-math major students. Primary responsibility includes conducting weekly recitation sections, office hours and grading homework.

Teaching Assistant for Pre-First Year Program, University of Pennsylvania*Summer 2023, 2024.*

Teaching assistant for UPenn PFP Math E (Vector calculus) mini-course to prepare incoming freshman for their future math courses. Responsibilities included holding office hours and grading.

SERVICE

Mentored undergraduate students via the Directed Reading Program at University of Pennsylvania on the following topics:

*The fundamental group**Spring 2026**Hilbert schemes**Fall 2025**Complex algebraic geometry**Fall 2024, Spring 2025**Scheme theory**Spring 2024**Tropical curves and Riemann-Roch**Fall 2023**Algebraic geometry, a first course**Summer 2023**Introduction to abstract varieties**Spring 2023*

FELLOWSHIPS & AWARDS

Benjamin Franklin & Fontaine Fellowship, University of Pennsylvania, 2022-2027.

Bridge-to-PhD program Fellowship, University of Pennsylvania, 2020-2022.

Undergraduate Dean's list-College of Engineering, Wayne State University. 2013-2017.

Undergraduate Research Opportunity Program (UROP) Creative Project Award, Wayne State University, May 2015.

Engineering Dean Discretionary Fund for a creative project, Wayne State University, Feb. 2015.

Tau beta Pi Michigan Epsilon Membership, Wayne State University, 2015 - 2020.

WORK EXPERIENCE

Automotive Engineer, General Motors, Detroit, MI

May 2017 - Aug. 2020

Worked as an engineer on several projects over the span of three years. Primary work involved developing Multi-Disciplinary Optimization (MDO) tools that would help in optimizing vehicle design with the target to achieve maximum occupant comfort.

REFERENCE

Available upon request.